

Canadian Mortgage Renewal Risk Project

Project Overview — Goal, Discovery Process & Findings

1. The Goal

I wanted a portfolio project that would be immediately relevant in an interview for an entry-level analyst role at a Canadian bank — something built with real institutional data, not a generic Kaggle dataset, and something I could speak to confidently for five minutes if asked "walk me through a project you're proud of." The goal was to combine three skills banks screen for at the analyst level: pulling data programmatically (Python), structuring and querying it properly (SQL), and turning it into a business insight (analysis + visualization).

2. How I Came Across This Problem

Rather than picking a topic arbitrarily, I worked backward from a question: what is a Canadian bank's retail/credit team actually worried about right now? The answer that kept surfacing was **mortgage renewals**. Canada has an unusually large share of mortgages that reset on 5-year fixed terms, and the Bank of Canada's overnight rate moved further and faster between 2022 and 2025 than at almost any point in the last two decades — from 0.25% up to 5.00%, then back down to 2.25%. That means a huge cohort of borrowers who signed 5-year mortgages during the ultra-low-rate pandemic years are renewing into a completely different rate environment than the one they locked in under, which directly affects delinquency risk, household spending, and loan portfolio quality — all things a credit or risk analyst would actually monitor.

That gave me a scoped, explainable problem: quantify the size of the "renewal shock" facing different borrower cohorts, using the Bank of Canada's own published rate data as the source of truth.

3. Approach

- **Extract:** Pull the Bank of Canada's target overnight rate history directly from their free, public Valet API (no key required) — the same data source banks' own economists reference.
- **Store:** Load it into a SQLite database with a proper schema: a fact table of rate decisions, a reference table of sample mortgage scenarios, and a derived table of calculated outcomes.
- **Analyze:** Write SQL using window functions to detect rate-cycle turning points, rolling averages, and pace-of-change — the kind of query an analyst is asked to write in a SQL screen.
- **Model:** In Python, apply the actual Canadian mortgage payment formula (semi-annual compounding) to five illustrative borrower scenarios who signed at different points in the cycle, and calculate their real monthly payment change at renewal.

- **Visualize:** Chart the rate cycle itself and the resulting renewal shock by scenario.

4. What I Discovered

The headline finding surprised me, and it's the kind of nuance that's worth bringing up in an interview: renewal outcomes right now are **not uniformly bad**. Whether a mortgage renewal is painful or a relief depends almost entirely on *when* the borrower originally signed — not on a single "rates are high" narrative.

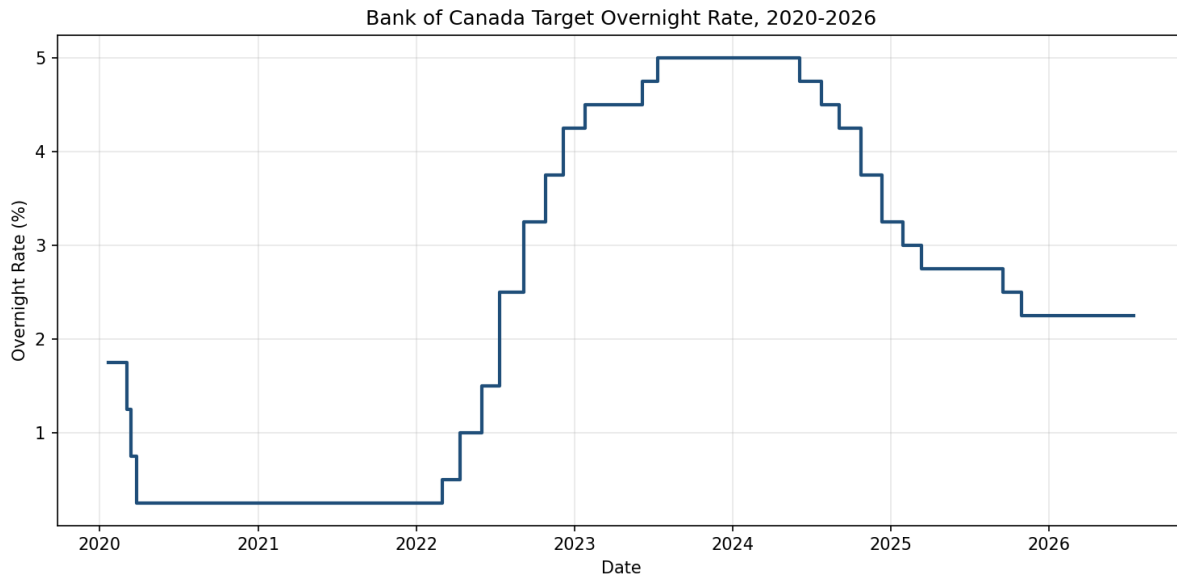


Figure 1. Bank of Canada target overnight rate, 2020–2026 — sourced from the Valet API (series V39079). Rates fell to 0.25% during the pandemic, rose to 5.00% by mid-2023 to fight inflation, then eased back to 2.25% through late 2025.

I modeled five illustrative borrowers who each signed a 5-year fixed mortgage at a different point in that cycle, then calculated their monthly payment change assuming they renew today into a ~4.29% fixed rate:

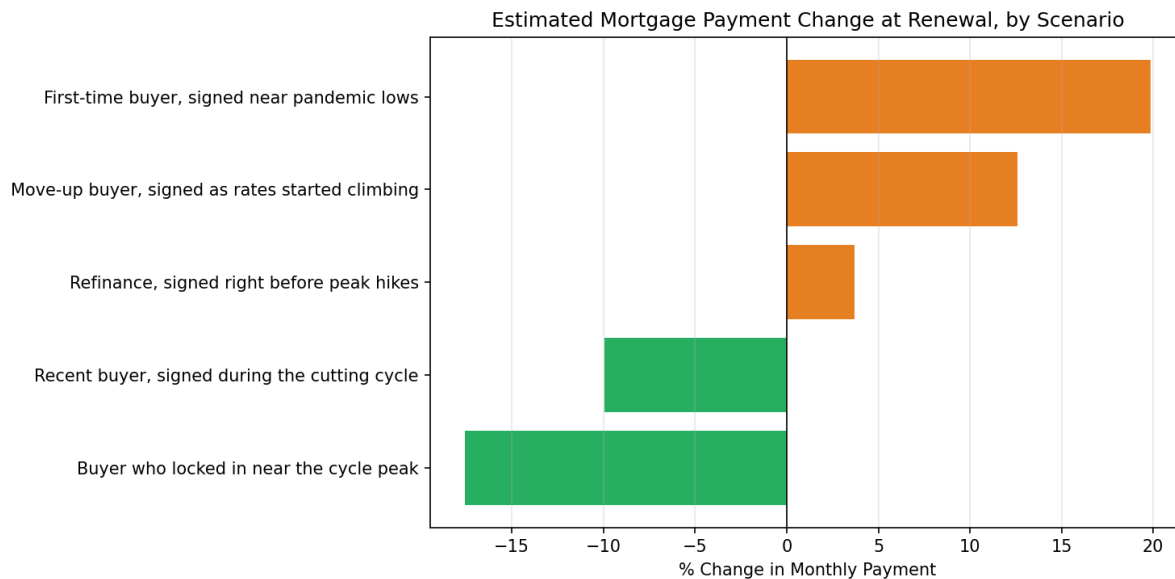


Figure 2. Estimated % change in monthly mortgage payment at renewal, by when the borrower originally signed. Positive = payment increase ("shock"), negative = payment relief.

Key takeaways:

- **Pandemic-era borrowers face the sharpest shock.** Someone who signed near the 2020 rate floor (~1.79%) is renewing into an estimated ~20% higher monthly payment — the segment a bank's risk team should be watching most closely for payment stress.
- **2023-vintage borrowers are actually getting relief.** Someone who locked in near the 5.00% peak is now renewing into a lower rate, an estimated ~18% payment *decrease* — the mirror image of the pandemic cohort.
- **The risk isn't evenly distributed across the loan book — it's cohort-specific.** A single portfolio-wide average would hide this entirely; the origination date is what actually predicts renewal risk, which is exactly the kind of segmentation a credit risk team performs.

5. Why This Matters for the Role

This project let me practice the exact workflow an analyst uses day to day — pull data from an external source, model it correctly in SQL, and translate it into a business-relevant conclusion — while producing an insight that's genuinely relevant to how Canadian banks think about credit risk on their residential mortgage books today.

Data source: Bank of Canada Valet API (bankofcanada.ca/valet). Mortgage scenarios are illustrative, not real borrowers. See [Step_by_Step_Guide.pdf](#) for how to reproduce or extend this project.